

Technical **Analytics** & Data Infrastructure

An Engineering Deep Dive into Namma Yatri Operational Efficiency and System Performance Metrics

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01_CORE_INFRASTRUCTURE

Pipeline & ETL Workflows

Analytics Tech Stack

COMPUTE_ENGINE

Data Modeling

SQL-driven data transformations for unified performance views. Advanced DAX implementations for dynamic metric filtering and context-aware insights.

VISUALIZATION_LAYER

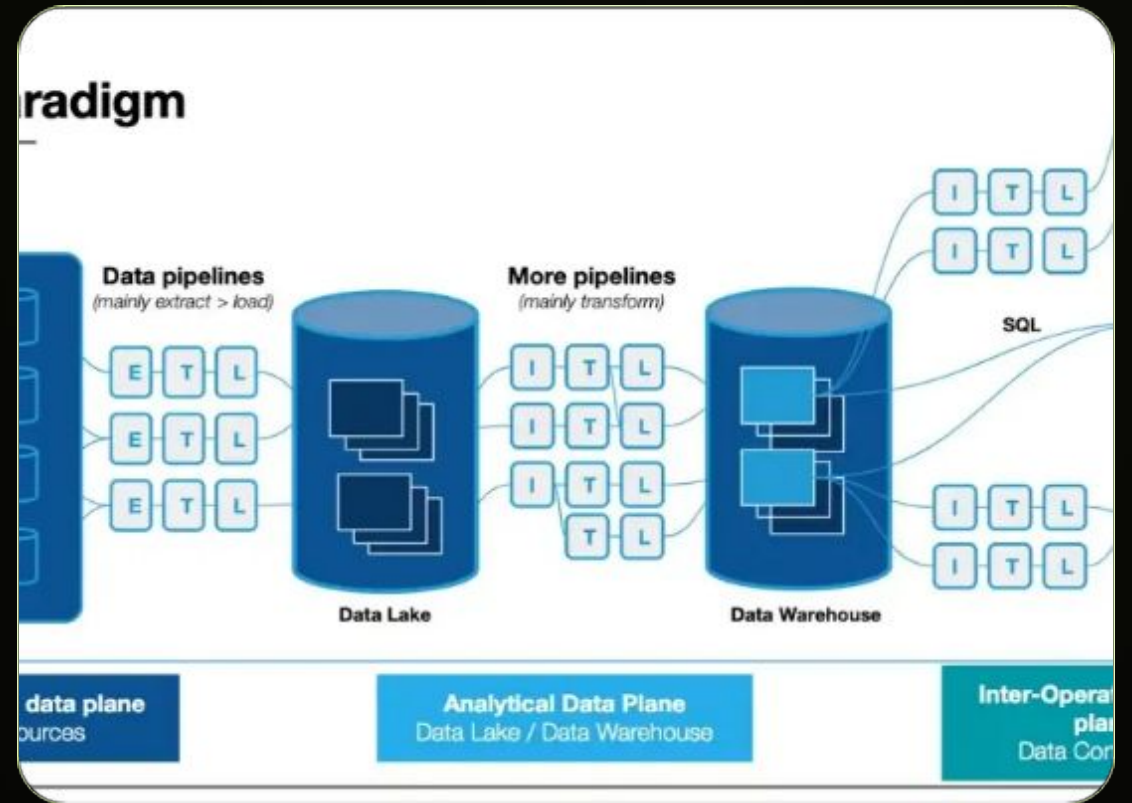
BI Architecture

Power BI and Tableau integration for real-time geospatial heatmaps and funnel analysis. Field parameters used for dynamic metric switching.

System Data Flow

The system utilizes an event-driven architecture to capture user intent from the mobile client to the cloud warehouse.

- **Ingestion:** Real-time streaming of 'Search' events.
- **Processing:** Matching algorithms calculate 'Estimates' and 'Quotes' in <200ms.
- **Persistence:** Transactional logs stored in high-concurrency databases for post-trip analytics.



Funnel Efficiency Logic

45.5%

Net Completion Rate

Probabilistic Modeling

Conversion is modeled as a product of conditional probabilities across the session lifecycle:

$$P(\text{Trip}) = P(\text{Search}) \times P(\text{Quote} | \text{Search}) \times P(\text{Trip} | \text{Quote})$$

GEOSPATIAL_ANALYSIS

Clustering Heatmaps

Advanced spatial clustering identifies high-friction zones.

Mahadevapura shows a cancellation variance of +12% compared to the mean.

Ramanagara remains the primary 'Search' cluster, requiring higher supply-node density to minimize the 40.91% abandonment rate.

Temporal Distribution (12-1 AM)



Analysis confirms a critical capacity strain during the midnight window, impacting the 40.91% abandonment metric.

Data Dictionary & Schema

Variable	Type	Definition	Business Logic
search_id	UUID	Unique session identifier	Primary key for funnel tracking
quote_status	Boolean	System response validity	Identifies supply coverage gaps
abandon_rate	Float	(1 - Completion/Search)	Measures friction-induced leakage
opp_gap_idx	Dec	Potential / Actualized	Scales market capture strategy

Systemic Challenges



Latency Gaps

33% Search-to-Quote drop suggests localized timeout or no-supply events in the matching engine.



Reliability

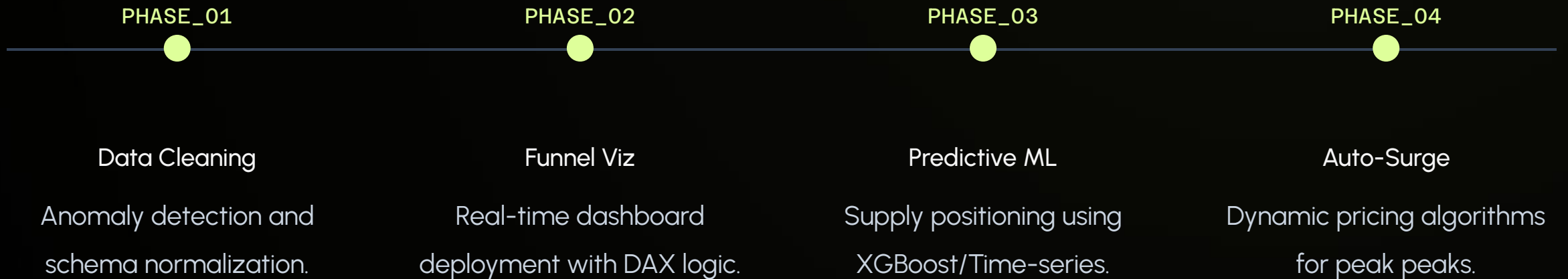
Combined cancellation rate of 13.61% points to supply-side churn in high-traffic corridors.



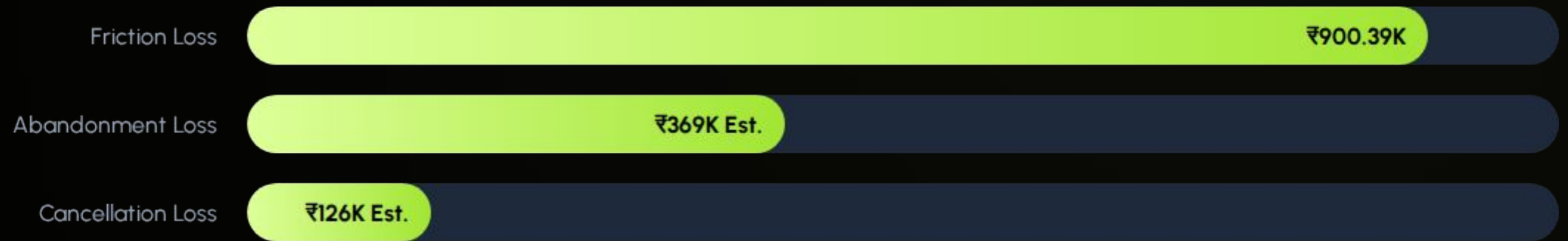
Gateway Health

Balanced 25% distribution across 4 methods requires high API availability to prevent final step drop-offs.

Engineering Roadmap



Financial Impact Quantification



Loss logic calculated via potential trip value extrapolation:

$$\text{Revenue Loss} = \sum (\text{Searches}_{\text{unfulfilled}} \times \text{AvgFare}_{\text{zone}})$$

Technical Q&A

Namma Yatri Engineering & Analytics Team

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